# EHR and MIMIC Database

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# Outline

- Introduction to HER
- Intensive Care Unit (ICU) Data
- Introduction to MIMIC database
- References

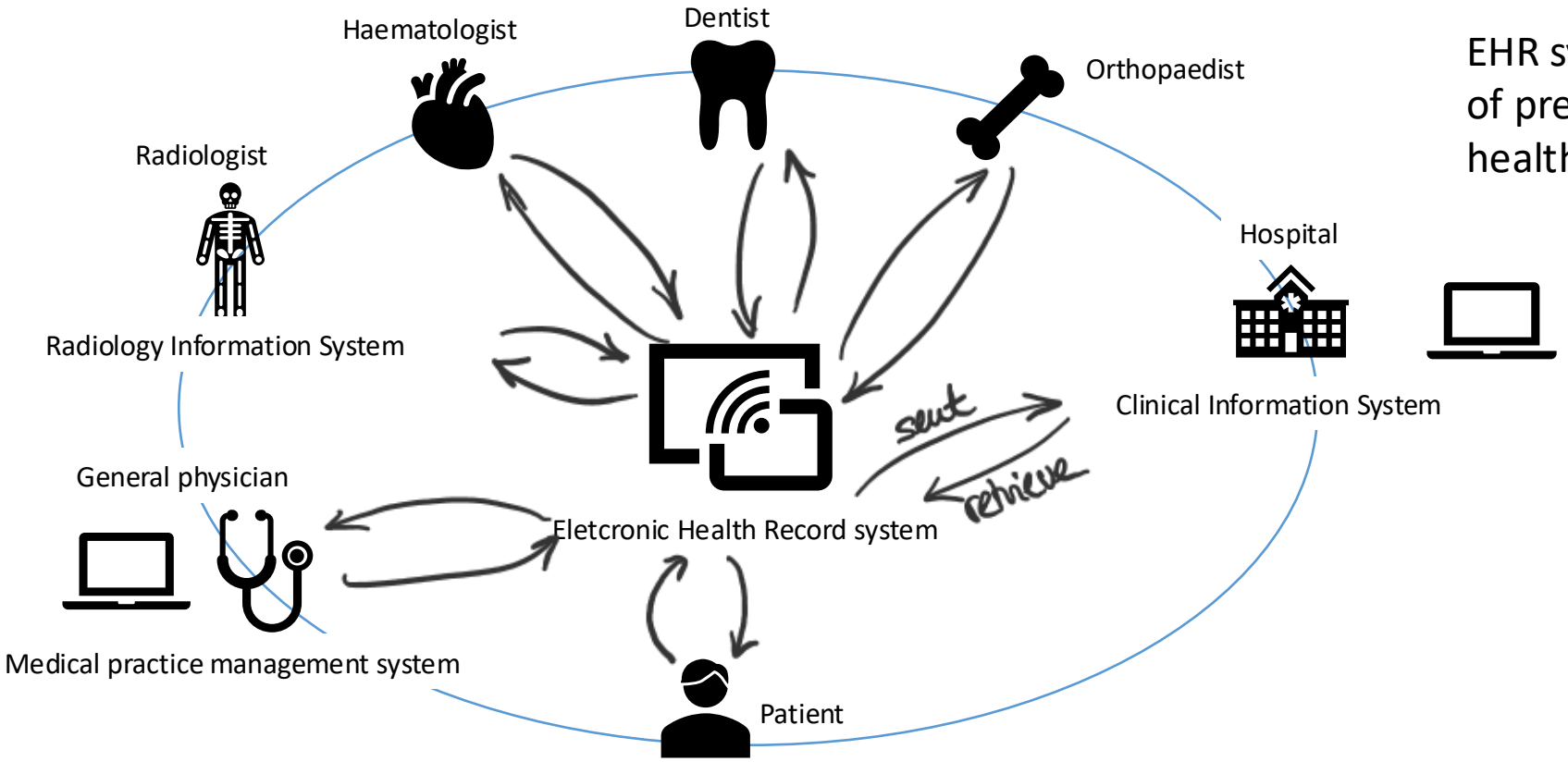
# Electronic health records (EHR)

# Challenges in Healthcare Before EHR Adoption

## Why electronic health records?

- Lack of cooperation and communication between different health care institutions
  - **Excess or shortage of care provision**. For example, a test is performed, even though another medical professional has already done this.
- Insufficient flow of information.
  - For example, if a person has an accident and, being unconscious, is unable to explain his state of health.
- Intransparent flow of information.
  - Passive role of patients. For example, some information that is passed on between different actors in the health care system is not available to patients.

# EHR system as a hub between healthcare institutions and related services



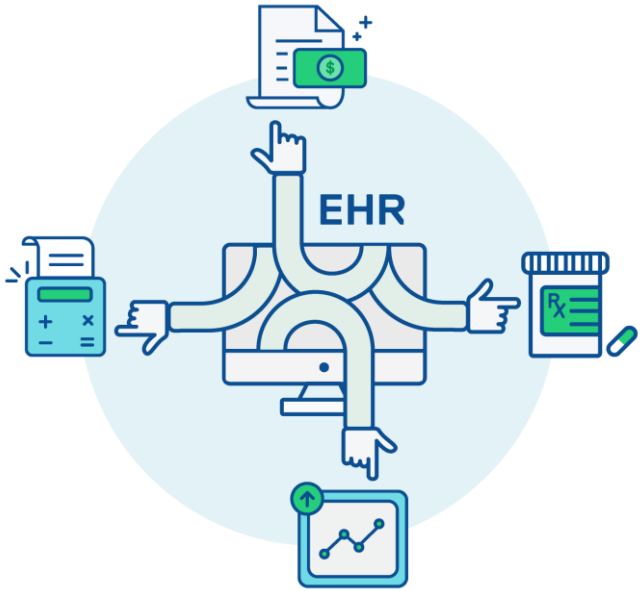

EHR system allows the linking of previously independent health information systems.

more responsive healthcare experience for patients

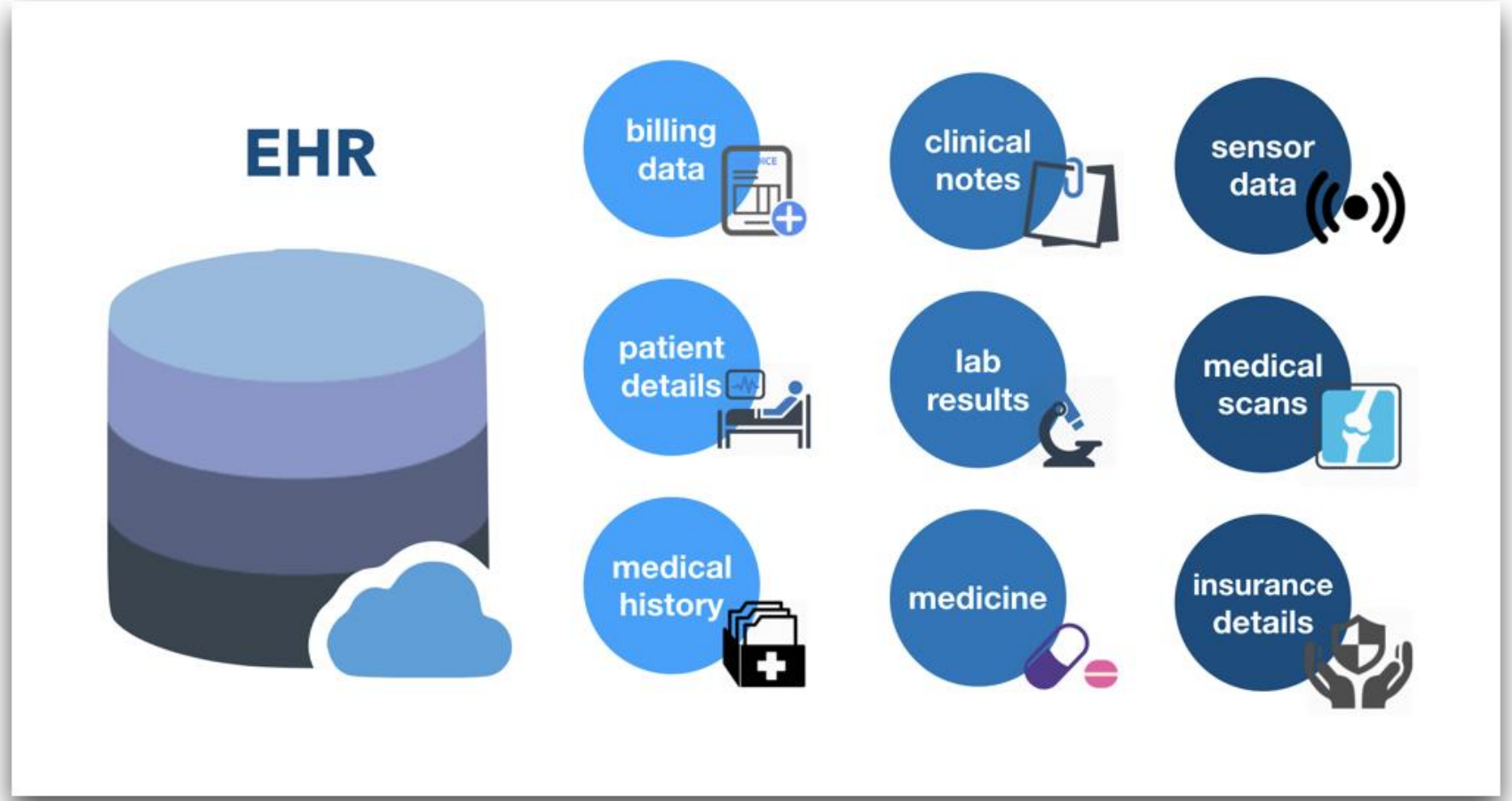
# Terminology

- **Electronic medical record (EMR):** sometimes called an Electronic Patient Record – is a **computerized medical record for individuals created in a service or an organisation that delivers care**, e.g. hospital, physician's office or long-term care facility. EMRs are typically **unique** to the provider or organisation, and allow storage, retrieval and modification of its patient records. (OECD, 2019)

- **Electronic health record (EHR):** An electronic health record (EHR) is a **longitudinal digital record that contains a history of all contact with the health system for an individual regardless of the settings, service and organisation** at which the contact took place. This can be achieved by either having one EHR across all settings and organisations in a health system ('one patient – one record') or provide a platform that can link multiple EMRs so that the information can be shared between settings and relevant agencies that use the data. (OECD, 2019)



# Data in EHR systems



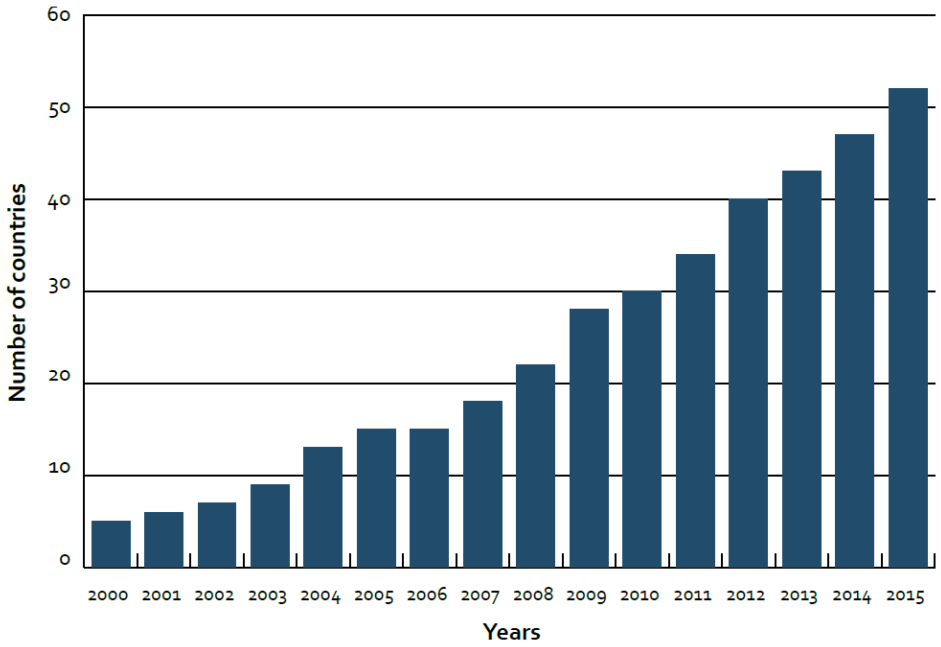

EHR systems can have data from a variety of different sources.

# Benefits of EHR adoption

- Improving quality of care
  - New forms of **cooperation** and easier **communication** between health actors.
  - Tracking who has viewed the information, increasing **transparency and reliability** of the information provided.
- Reducing healthcare costs
  - Reducing redundant diagnostic tests.
  - Reducing the number of malpractice claims.
- Facilitating patient empowerment
  - Access to health information systems .
- Supporting health research and policy-making
  - Source of information for big data analytics.

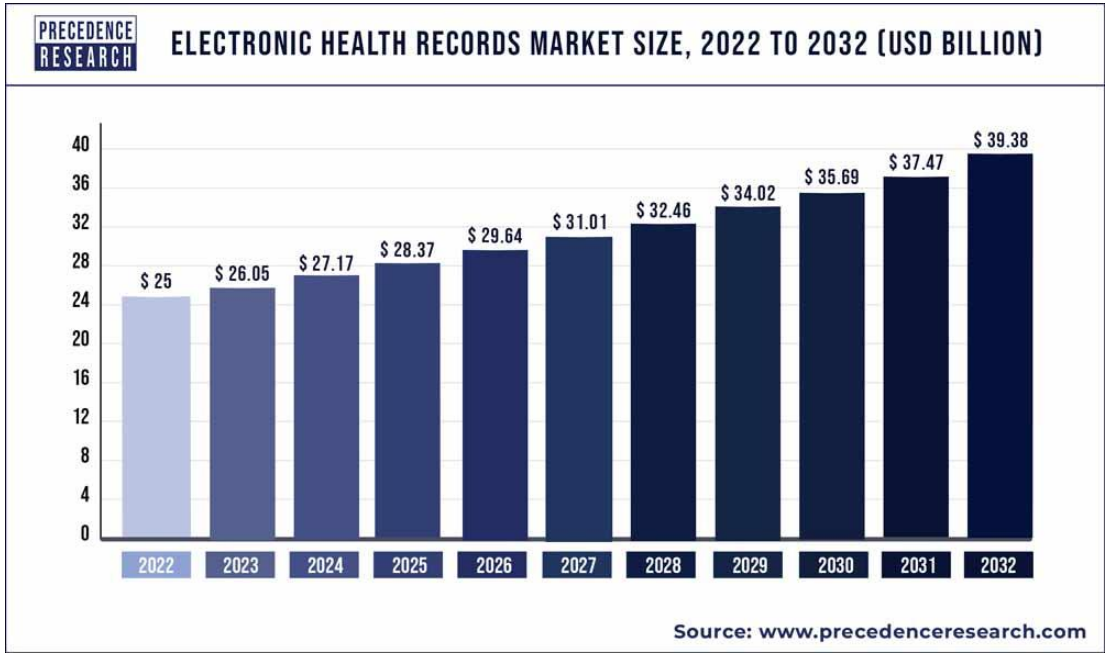
# EHR Adoption and Market

**Figure**: Countries with **national** EHR system, cumulatively by year of adoption (2000–2015).




a **Note:** This figure does not show the complete picture, as not all countries reported their year of adoption.

Reprinted from Global diffusion of eHealth: making universal health coverage achievable. Report of the third global survey on eHealth. Geneva: World Health Organization; 2016. Licence: CC BY-NC-SA 3.0 IGO



# But, are EHRs really a success?

- When it comes to adoption of EHRs, the initiatives were a success:
  - Presently more than 90% of hospitals and close to nine in 10 office-based physicians have adopted EHRs.
- However, EHRs have encounter many difficulties.
  - Unmet potential of EHRs: meaningful use and health information exchange.
- Health system leaders often agree.

  "*We have made a colossal transformation in a relatively short period of time*,"

  "*But we have not realized the potential benefits of EHRs*."

  Lloyd Minor, MD, Dean of the Stanford School of Medicine at a 2018 symposium.

Source: "The EHR Is Dead. Long Live The EHR Platform". Forbes 2021.

# Challenges of EHR (I)

1. Data protection issues
   - How to protect privacy and access to the data
   - Healthcare data is highly regulated. Privacy and security concerns are paramount.

2. Implementation and operational costs
   - Changes in clinical workflows lead to temporary loss of productivity.
   - EHRs are a leading cause of physician burnout
   - EHR vendors have been sued for selling non functioning products to hospitals and accused of acts of fraud.

# Challenges of EHR (II)

## 3. Interoperability between health systems, technologies and devices

**Interoperability in healthcare:** "[…] the ability of different information technology systems and software applications to communicate, exchange data, and use the information that has been exchanged." (Healthcare Information and Management Systems Society, Inc. HIMSS, 2013)

- Data exchange **standards** and schema are designed to permit data to be shared between different healthcare stakeholders regardless of the application or its vendor.

ISO 13606

HL7® FHIR®

SMART®

open EHR

EHDEN EUROPEAN HEALTH DATA & EVIDENCE NETWORK

# Challenges of EHR (III): Interoperability

Standards exist, but inconsistent adoption and implementation still limit true health data interoperability.

- **ISO 13606** is a standard developed by the **International Organization for Standardization** for communicating parts or all of an electronic health record between EHR systems or between EHR systems and a centralized EHR data repository. It is also applicable for communication between EHR systems or EHR repositories and clinical applications, other middle ware components (e.g. decision support components), personal health applications/devices, or as the representation of EHR data within a distributed record system. http://www.en13606.org/information.html

ISO 13606

- Founded in 1987, the **Health Level Seven International** (**HL7**) is a global not-for-profit standards development organization that has developed a generic model which have served as a foundation for EHR systems. The HL7 Clinical Document Architecture (**CDA**) is an XML-based markup standard intended to specify the encoding, structure and semantics of clinical documents for exchange. https://wiki.hl7.org

- HL7 has created **Fast healthcare Interoperability Resources** (**FHIR**), a standard describing data formats and elements for the exchange of EHR. Utilizes application programming interfaces (APIs) to facilitate **health data exchange**. Provides standardized **Resources** (information unit). http://www.hl7.org/fhir/

- In this context, **Substitutable Medical Applications, Reusable Technologies** (**SMART)** on FHIR is an open standards-based platform, run by the SMART Health IT project from the Boston Children's Hospital Computational Health Informatics Prorgram, enabling connecting EHR systems to medical apps. https://smarthealthit.org

# Challenges of EHR (IV): Interoperability

- **OpenEHR** is a global initiative that developed an open standard and an open-source software for implementation. The various artefacts of the initiative are produced by the openEHR community and managed by openEHR International, an international non-profit organization. openEHR is optimized to provide a **data platform with a stronger focus on data persistence.** https://openehr.org/

- The **European Health Data and Evidence Network** (**EHDEN**) is striving to develop a common data model to facilitate health statistics, monitoring and research undertaken by various players in the healthcare environment (e.g. governments, private sector entities, private sector entities such as pharmaceutical research etc.) approach for the standardization of health data. https://www.ehden.eu

# Challenges of EHR (V)

## 4. Design

- The EHR systems have not been designed considering the needs of healthcare professionals and patients.
- **Intuitive user interface** is essential when it comes to avoiding medication errors,
  - Poorly designed interfaces can even increase the risk of errors.
- Success factors: **People-centered design** and promote **usability**

# EHR in Catalunya

**Historial Electrònic de Salut (HES)**

- HC3 is the electronic history that groups the set of documents containing relevant data and information about the situation and evolution of a patient throughout his/her care process.
- It allows the access and consultation in an immediate, safe and confidential way of the relevant information available on the patients in the different sanitary centers of the network of public assistance of Catalunya.
  - But for the moment, not for private services and not for the rest of Spain….

La meva
/Salut

# Intensive Care Unit (ICU) Data

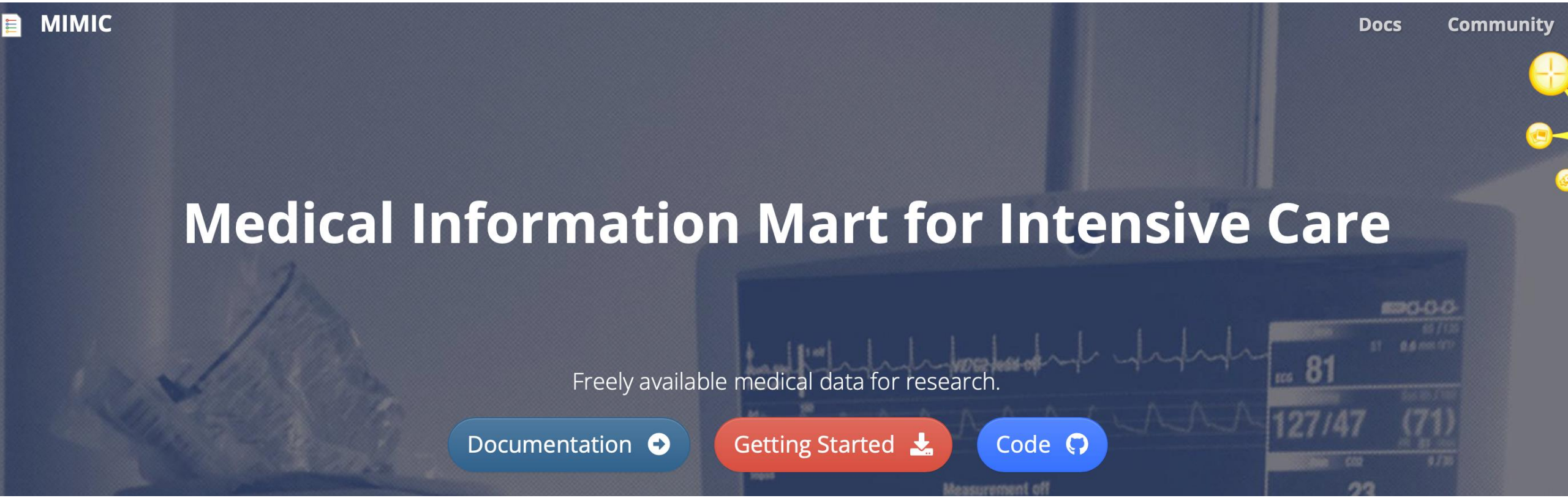
# ICU Data: A Compressed, High-Intensity EHR

- The ICU is an especially data-rich environment as patients require close monitoring.
- **Compressed Timeline**
  - Outpatient care → months/years
  - ICU care → hours/days
  - Observe full trajectory: deterioration → treatment → outcome
- **High-Intensity Signals**
  - Vitals: every few minutes
  - Labs: every few hours
  - Continuous treatments & rapid physiological changes
- **Full EHR Ecosystem**
  - Demographics, diagnoses, labs, medications, procedures, notes, time-series vitals

# ICU Public Databases

- MIMIC Database
- eICU Collaborative Research Database
- **AmsterdamUMCdb**
- HiRID (High Time Resolution ICU Dataset)
- The Pediatric Intensive Care (PIC) database

| Dataset | Region | Multi-Center | High-Frequency | Imaging | Specialty Focus |
|---|---|---|---|---|---|
| MIMIC-IV | US | No | Moderate | Yes (via MIMIC-CXR) | General ICU |
| eICU | US | Yes | Moderate | No | General ICU |
| AmsterdamUMCdb | NL | No | Moderate | Limited | General ICU |
| HiRID | CH | No | Very High | No | Physiology |
| AUMC COVID | NL | No | Moderate | No | COVID ICU |




# MIMIC Database

# MIMIC Database

- MIMIC (Medical Information Mart for Intensive Care) is a large, freely-available database comprising deidentified health-related data from patients who were admitted to the critical care units of the Beth Israel Deaconess Medical Center. https://mimic.mit.edu
- The database includes information such as:
  - demographics
  - vital sign measurements made at the bedside
  - laboratory test results
  - procedures
  - medications
  - caregiver notes
  - imaging reports
  - mortality

# MIMIC Versions

- **MIMIC-IV** (relased in 2023) contains data from 2008-2019. The data was collected from the ICU information system (Metavision) bedside monitors. https://mimic.mit.edu/docs/iv/

- **MIMIC-III** (relased in 2016) contains data from 2001-2012. The data was collected from Metavision and CareVue bedside monitors. https://mimic.mit.edu/docs/iii/

- **MIMIC-II** (relased in 2011) contains data from 2001-2008. The data was collected from CareVue bedside monitors. MIMIC-II is no longer publicly available but the data can still be obtained from MIMIC-III by only including data from the CareVue monitors.

# MIMIC-IV Database

- MIMIC-IV contains data for over 65,000 patients admitted to an **ICU** and over 200,000 patients admitted to the **emergency department**, from 2008-2019. The data was collected from Metavision bedside monitors.

- The MIMIC-IV database includes information such as:
  - demographics
  - vital sign measurements made at the bedside
  - laboratory test results
  - procedures
  - medications
  - caregiver notes
  - imaging reports
  - mortality
  - Chest x-ray data

scientific **data**

Check for updates

OPEN

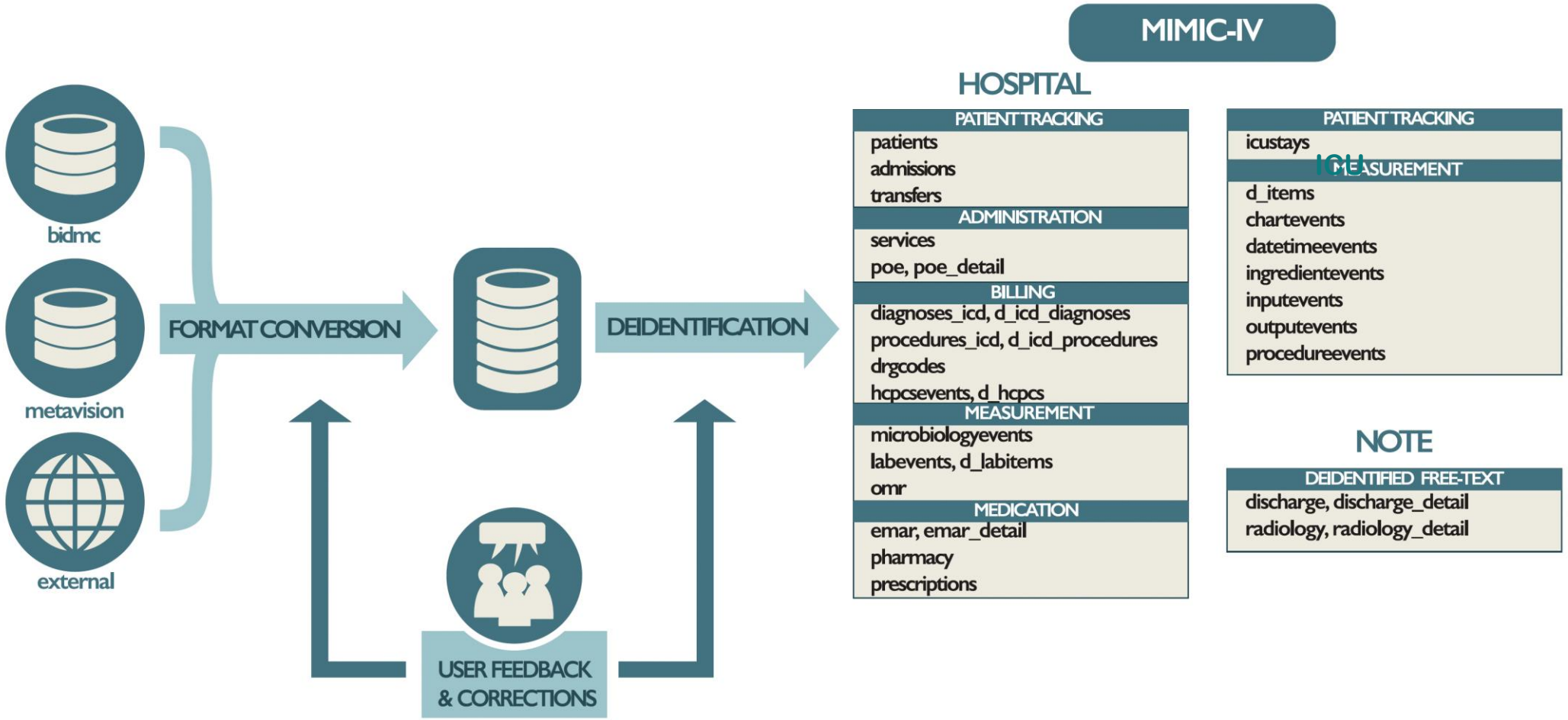
DATA DESCRIPTOR

## MIMIC-IV, a freely accessible electronic health record dataset

Alistair E. W. Johnson[1,2] ✉, Lucas Bulgarelli[1], Lu Shen[3], Alvin Gayles[3], Ayad Shammout[3], Steven Horng[3], Tom J. Pollard[1], Sicheng Hao[1], Benjamin Moody[1], Brian Gow[1], Li-wei H. Lehman[1], Leo A. Celi[1,3] & Roger G. Mark[1]

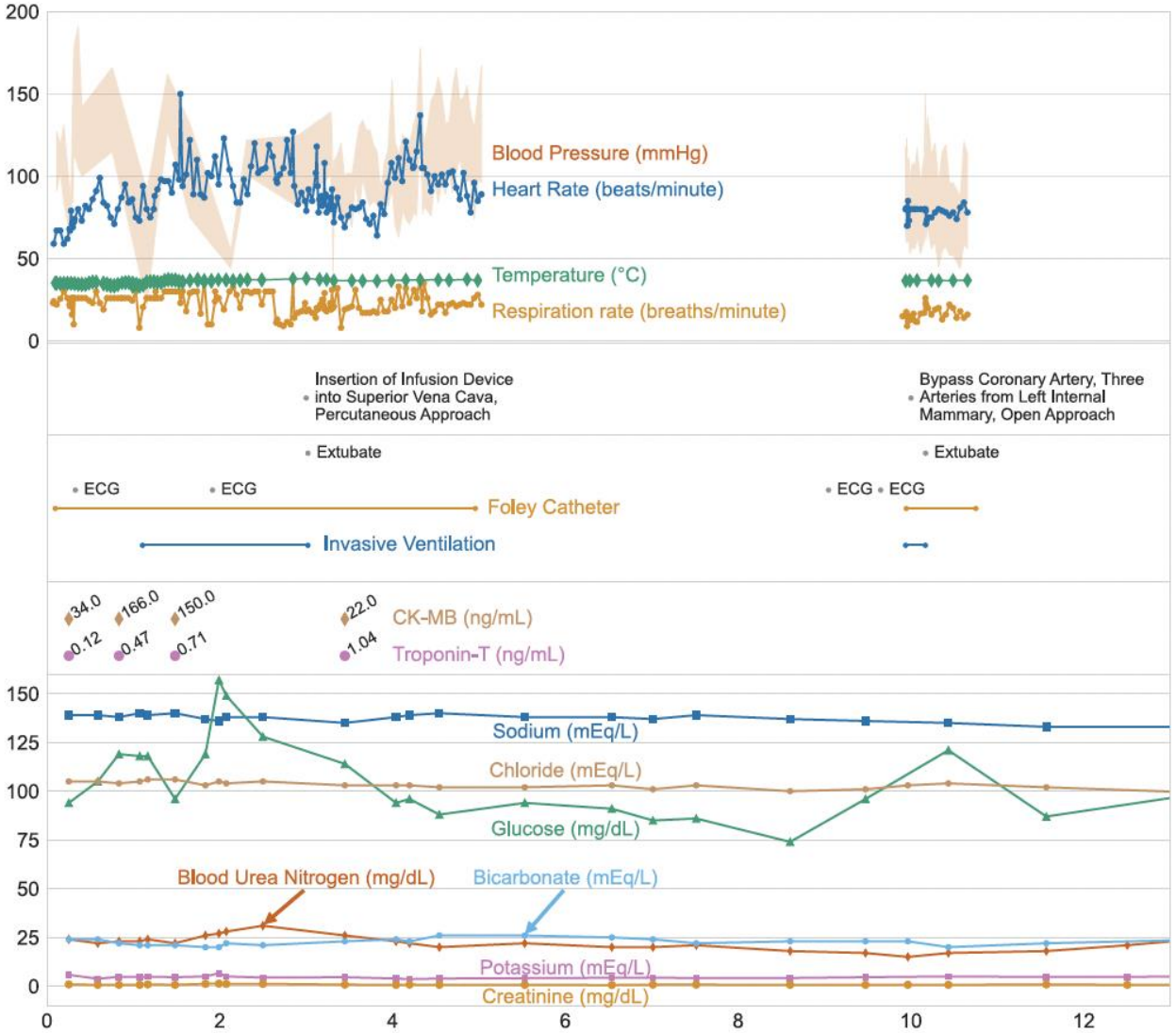
Scientific Data | (2023) 10:1 | https://doi.org/10.1038/s41597-022-01899-x

# MIMIC-IV






Johnson, A.E.W., Bulgarelli, L., Shen, L. *et al.* MIMIC-IV, a freely accessible electronic health record dataset. *Sci Data* **10**, 1 (2023). https://doi.org/10.1038/s41597-022-01899-x




Johnson, A.E.W., Bulgarelli, L., Shen, L. *et al*. MIMIC-IV, a freely accessible electronic health record dataset. *Sci Data* **10**, 1 (2023). https://doi.org/10.1038/s41597-022-01899-x

**Fig. 3** Visualization of data within MIMIC-IV for a single patient's hospitalization: `hadm_id 28503629`. Three vertically stacked panels highlight the variety of information available. Vital signs are shown in the top panel: note the frequency of data collection for temperature is much higher at the start of the ICU stay due to the use of targeted temperature management. Procedures from multiple sources are shown in the middle panel, including from billing information, the provider order entry system, as well as the ICU information system. The bottom panel displays patient laboratory measurements. Note that while frequent vital signs are only available when the patient is in the ICU, laboratory measures are available throughout their hospitalization.

# What's new in MIMIC-IV

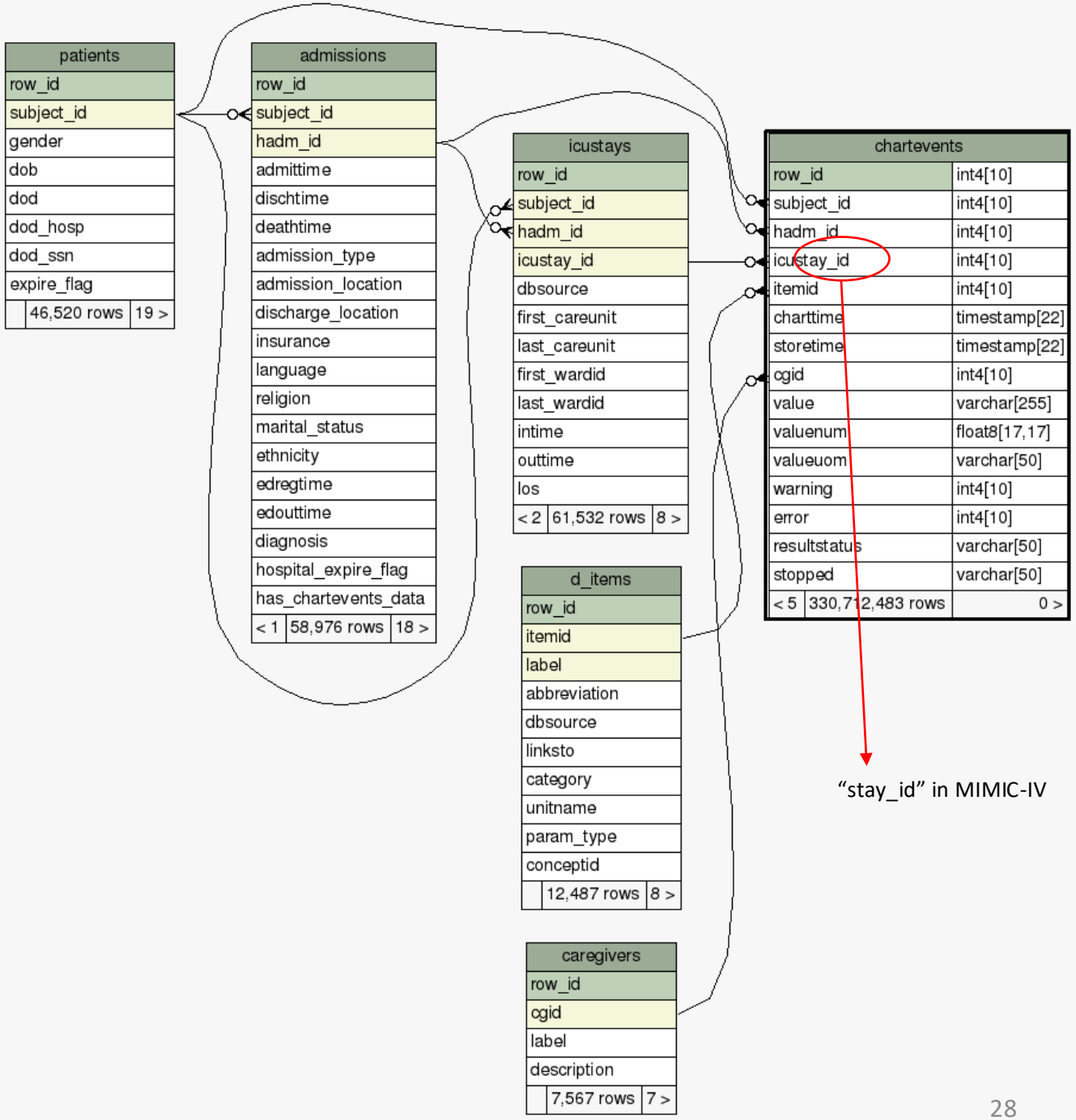
- More recent data (2008-2019)
- Changes in the structure (**Modules**)
  - HOSP: data acquired from the hospital wide EHR.
  - ICU: information collected from the clinical information system used within the ICU (MetaVision).
  - ED: data collected while patients are in the Emergency Department.
  - NOTE: deidentified free-text clinical notes for hospitalized patients.
  - CXR: lookup tables linking patient identifiers with MIMIC-CXR images
- Adding new data elements
  - Imaging data (Chest x-ray data)
  - Data sourced from the relatively newly installed electronic Medicine Administration Record (eMAR) system. Bedside staff will scan barcodes for each individual formulary unit of a medication when administering it.

# MIMIC-IV Database

- Some of the tables in ICU module contains events:
  - patient outputs (OUTPUTEVENTS table)
  - procedures (PROCEDUREEVENTS table)
  - information documented as a date or time (DATETIMEEVENTS table)
  - charted information (CHARTEVENTS table)
- All events tables contain a *stay_id* column allowing identification of the associated ICU patient in icustays, and an *itemid* column allowing identification of the concept documented in table D_ITEMS.

- More info: https://mimic.mit.edu/docs/iv/modules/

# Relations of a subset of tables of the MIMIC-III database



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# MIMIC-IV Database

- CHARTEVENTS contains all the charted data available for a patient.
- Every row of CHARTEVENTS is associated with a single **itemid** which represents the **concept measured**, but it does not contain the actual name of the drug or test.
- D_ITEMS is a **dictionary table** with definitions for all items in the ICU databases: itemid
- By joining CHARTEVENTS and D_ITEMS it is possible to identify what concept a given ITEMID represents.

| | subject_id | hadm_id | stay_id | caregiver_id | charttime | storetime | itemid | value | valuenum | valueuom | warning |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **0** | 10005817 | 20626031 | 32604416 | 6770.0 | 2132-12-16 00:00:00 | 2132-12-15 23:45:00 | 225054 | On | NaN | NaN | 0.0 |
| **1** | 10005817 | 20626031 | 32604416 | 6770.0 | 2132-12-16 00:00:00 | 2132-12-15 23:43:00 | 223769 | 100 | 100.0 | % | 0.0 |
| **2** | 10005817 | 20626031 | 32604416 | 6770.0 | 2132-12-16 00:00:00 | 2132-12-15 23:47:00 | 223956 | Atrial demand | NaN | NaN | 0.0 |
| **3** | 10005817 | 20626031 | 32604416 | 6770.0 | 2132-12-16 00:00:00 | 2132-12-15 23:47:00 | 224866 | Yes | NaN | NaN | 0.0 |
| **4** | 10005817 | 20626031 | 32604416 | 6770.0 | 2132-12-16 00:00:00 | 2132-12-15 23:45:00 | 227341 | No | 0.0 | NaN | 0.0 |

# Censored labels in MIMIC-IV

- All dates have been deidentified by shifting them into a future time period between 2100 - 2200.
- The anchor_year column is a deidentified year occurring sometime between 2100 - 2200, and the anchor_year_group column is one of the following values: "2008 - 2010", "2011 - 2013", "2014 - 2016", "2017 - 2019", and "2020 - 2022".
- For example, if a patient's anchor_year is 2158, and their anchor_year_group is 2011 - 2013, then any hospitalizations for the patient occurring in the year 2158 actually occurred sometime between 2011 - 2013.

# Censored labels in MIMIC-IV

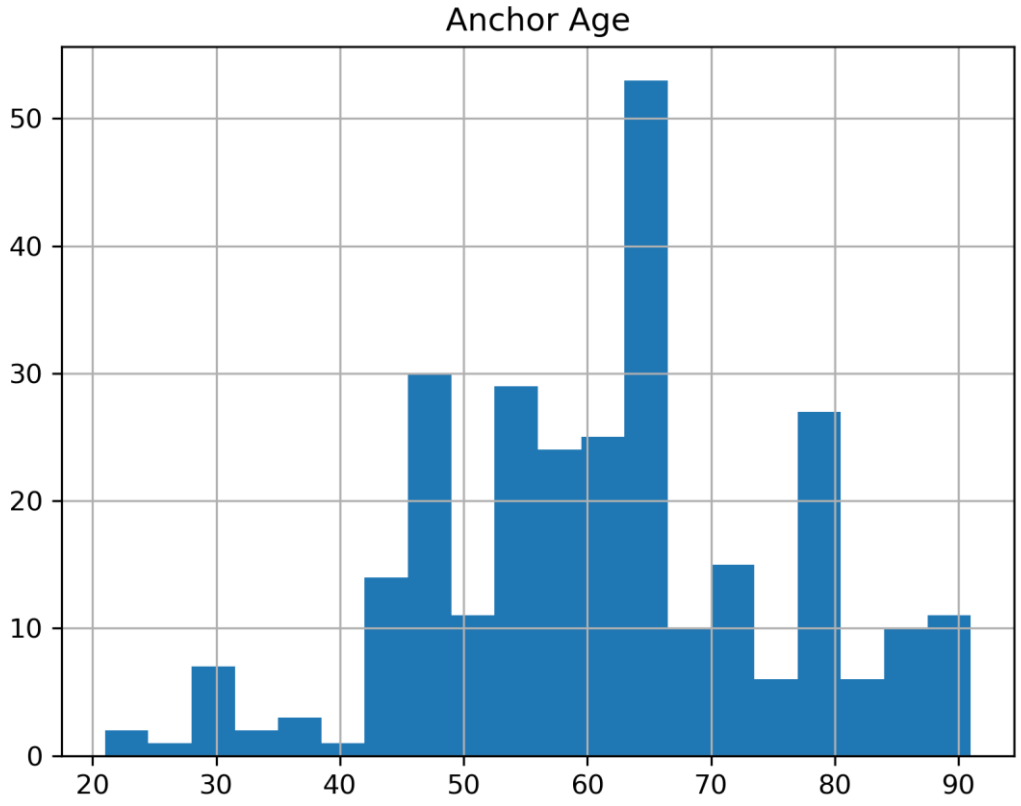
- Given that:
  - **anchor_year** is a shifted year for the patient.
  - **anchor_year_group** is a range of years - the patient's anchor_year occurred during this range.
  - **anchor_age** is the patient's age in the anchor_year.
- We can compute **the age of a patient at the time of the ICU admission**. Example:
  - A patient has an anchor_year of 2153, anchor_year_group of 2008 - 2010, and an anchor_age of 60.
    - The year 2153 for the patient corresponds to 2008, 2009, or 2010.
    - The patient was 60 in the shifted year of 2153, i.e. they were 60 in 2008, 2009, or 2010.
    - The ICU admission time is 2155-02-25 23:37:19
  - Therefore, the age of the patient at the time of the ICU admission is computed as

    **intime_age = ICU admission time - anchor_year + anchor_age**

    → ICU time age = 2155 – 2153 + 60 = **62**

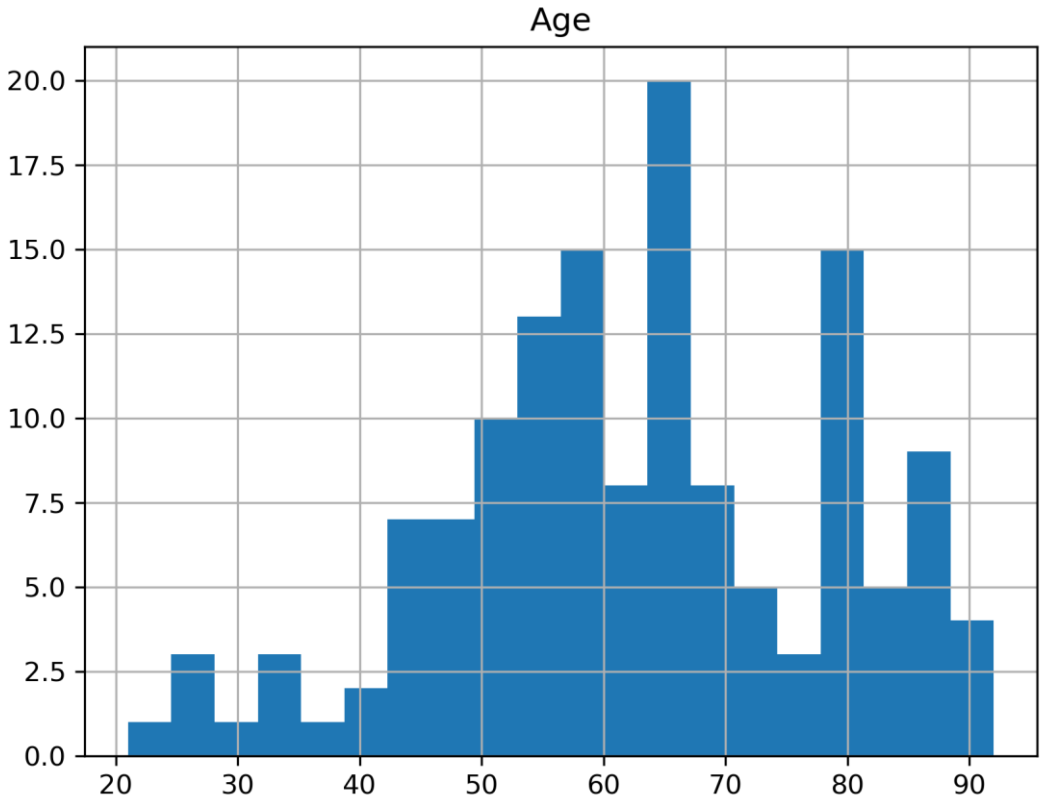
# Censored labels in MIMIC

- We show the anchor age of the Demo MIMIC database.



# Censored labels in MIMIC

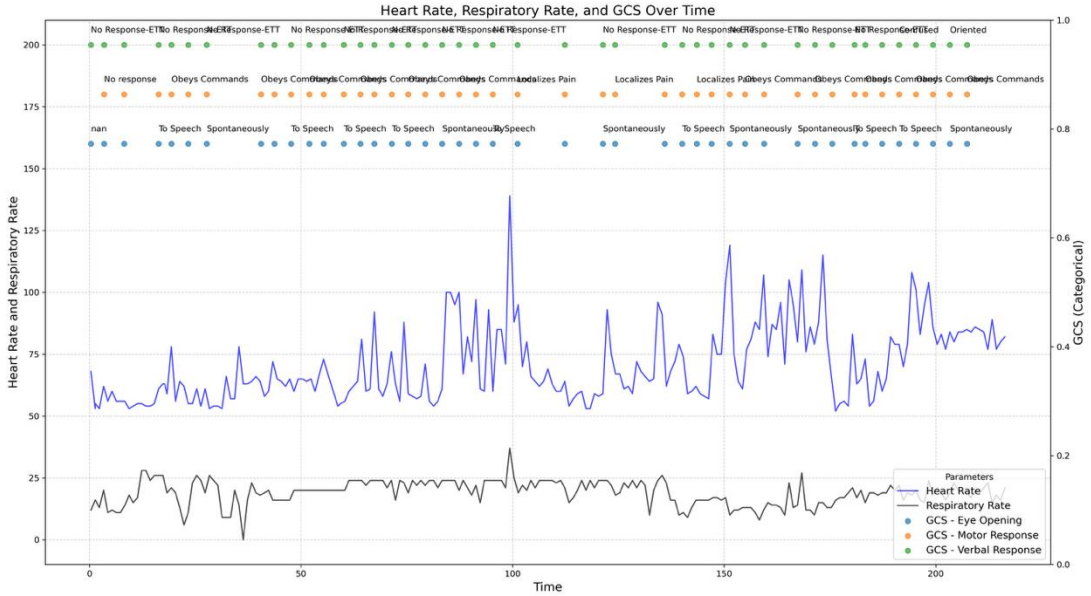
- We transform into age at the time of the ICU admission.



# Exemples and exercises

# Example and exercise on MIMIC-IV

- Download the notebooks from CV to use for your work.



# References

For more details about benefits of EHR adoption:


- Casey et al. *Using Electronic Health Records for Population Health Research: A Review of Methods and Applications*. Annu Rev Public Health. 2016;37:61-81.doi:10.1146/annurev-publhealth-032315-021353
- Campanella et al. *The impact of electronic health records on healthcare quality: a systematic review and meta-analysis*, European Journal of Public Health, Volume 26, Issue 1, February 2016, Pages 60–64, https://doi.org/10.1093/eurpub/ckv122
- Menachemi N, Collum TH. *Benefits and drawbacks of electronic health record systems*. Risk Manag Healthc Policy. 2011;4:47-55. doi:10.2147/RMHP.S12985
- Bertelsmann Report (https://www.bertelsmann-stiftung.de/fileadmin/files/Projekte/Der_digitale_Patient/VV_SHS-Studie_EN.pdf)

# References


- Benjamin Shickel, Patrick Tighe, Azra Bihorac, Parisa Rashidi. Deep EHR: A Survey of Recent Advances in Deep Learning Techniques for Electronic Health Record (EHR) Analysis. https://arxiv.org/abs/1706.03446
- Johnson, A., Pollard, T., Shen, L. *et al.* MIMIC-III, a freely accessible critical care database. *Sci Data* **3,** 160035 (2016). https://doi.org/10.1038/sdata.2016.35

# To complement today's class

- Why electronic health records? https://www.youtube.com/watch?v=Lo_3qOejQzI&feature=emb_imp_woyt
- Electronic Health Records: Usability and Unintended Safety Issues: https://www.youtube.com/watch?v=SlQ60KaLeHg
- Interoperability of Electronic Health Records– Benefits and Opportunities: https://www.youtube.com/watch?v=gWfzcOliRac
- European Health Information Gateway: https://gateway.euro.who.int/en/search/?q=EHR#page_indicators=1
- FHIR: https://www.youtube.com/watch?v=OIt0GrCPu8k

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